

CS333 Final Part (a)

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I. MULTIPLE CHOICE QUESTIONS (50 POINTS)

1. Given the time complexity equation for a recursive algorithm as $T(N) = 4T(N/2) + \mathcal{O}(N^2)$. What is the time complexity of that algorithm?

- a) $\mathcal{O}(N)$
- b) $\mathcal{O}(N \log N)$
- c) $\mathcal{O}(N^2)$
- d) $\mathcal{O}(N^2 \log N)$
- e) $\mathcal{O}(N^3)$

2. What is the value of *sum* after the following code fragment is executed?

```
int sum = 0;
for (i = N; i >= 1; i--)
    for (j = 2 * i; j <= 2 * N; j += 2)
        sum++;
```

- a) N
- b) $(N^2 + N)/2$
- c) N^2
- d) N^3
- e) N^4

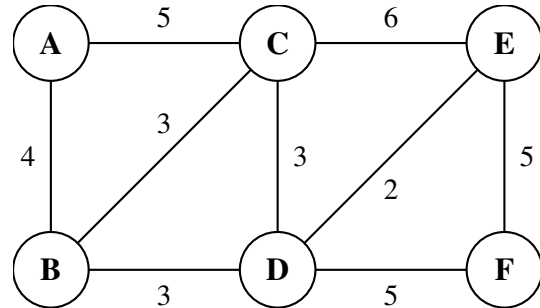
3. Which of the following functions has the largest growth?

- a) n^2
- b) n
- c) $n^4 \log(n)$
- d) 1.001^n
- e) n^{50}

4. What is the worst case time complexity of the algorithm (for a graph with N nodes) which checks if node i has no outgoing edges (i) adjacency matrix, (ii) adjacency list representation?

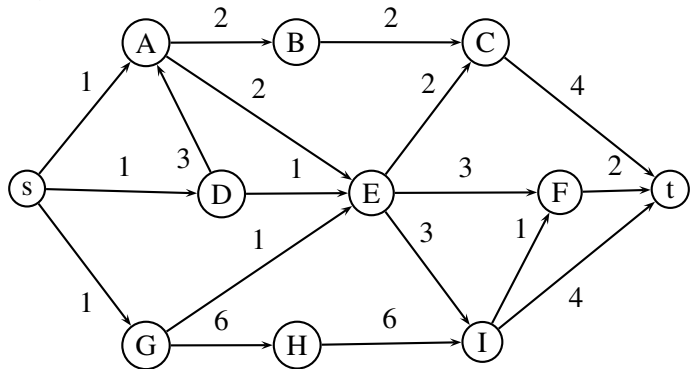
- a) $\mathcal{O}(\log N)$, $\mathcal{O}(\log N)$
- b) $\mathcal{O}(\log N)$, $\mathcal{O}(1)$
- c) $\mathcal{O}(N)$, $\mathcal{O}(1)$
- d) $\mathcal{O}(N)$, $\mathcal{O}(N^2)$
- e) $\mathcal{O}(1)$, $\mathcal{O}(N)$

5. Given the graph below, what is the length of the



minimum spanning tree?

- a) 14
- b) 15
- c) 16
- d) 17
- e) 18



6. In the graph above, what is the shortest path from s to t ?

- a) 11
- b) 10
- c) 9
- d) 8
- e) 7

7. Which of the following denominations can not be solved optimally by the greedy algorithm (Coin Change problem)?

- a) 1, 5, 10, 20
- b) 1, 10, 20, 100
- c) 1, 3, 6, 12
- d) 1, 4, 8, 16
- e) All of them can be solved

8. What is the minimum number of bits used to represent the string "abcd" according to Huffman encoding, if the number of occurrences of the characters are:

a = 2, b = 2, c = 2, d = 2.

- a) 6
- b) 8
- c) 10
- d) 12
- e) 14

9. Which of the following greedy strategy is optimal for interval partitioning problem? In interval partitioning problem, the goal is to find the minimum number of classrooms to schedule so that no two occur at the same time in the same room. In the end, all lectures must be scheduled.

- a) Lecture having the earliest start time is selected first
- b) Lecture having the earliest finish time is selected first
- c) Lecture having fewest conflict is selected first
- d) None of them are optimal
- e) All of them are optimal

10. If we want to obtain a leftly imbalanced binary search tree from numbers 1 to $2^n - 1$, what is the correct order of insertion of those numbers to the binary search tree?

- a) $2^n - 1, \dots, 2, 1$
- b) $1, 2, \dots, 2^n - 1$
- c) $2^{n-1}, 2^{n-2}, \dots, 1$
- d) $1, 3, 5, \dots, 2, 4, 6, \dots$
- e) $2^{n-1}, 2^{n-2}, 2^{n-1} + 2^{n-2}, \dots, 2^n - 1$

11. What is the average time complexity for insertion / search in a fully balanced binary search tree?

- a) $\mathcal{O}(1), \mathcal{O}(\log N)$
- b) $\mathcal{O}(1), \mathcal{O}(N)$
- c) $\mathcal{O}(1), \mathcal{O}(1)$
- d) $\mathcal{O}(\log N), \mathcal{O}(\log N)$
- e) $\mathcal{O}(\log N), \mathcal{O}(N)$

12. What is the average and worst case time complexity of Merge sort?

- a) $\mathcal{O}(N), \mathcal{O}(N)$
- b) $\mathcal{O}(N), \mathcal{O}(N \log N)$
- c) $\mathcal{O}(N \log N), \mathcal{O}(N \log N)$
- d) $\mathcal{O}(N \log N), \mathcal{O}(N^2)$
- e) $\mathcal{O}(N^2), \mathcal{O}(N^2)$

13. Algorithm B solves problems of size N by recursively solving sixteen subproblems of size $N/4$ and then combining the solutions in constant time. What is the time complexity of B?

- a) $\mathcal{O}(N^2 \log N)$
- b) $\mathcal{O}(N^2)$
- c) $\mathcal{O}(N^3)$
- d) $\mathcal{O}(N \log N)$
- e) $\mathcal{O}(N)$

14. In the multiplication of two n by n matrices, the standard divide and conquer approach divides matrices into 4 equal parts, and then it multiplies and adds these submatrices to get the end result. The number of multiplications required to get the result matrix is 8 and the internal matrices are added in $\mathcal{O}(N^2)$ time. Let say you found out a new algorithm with 5 multiplications instead of 8, then your new algorithm will have the time complexity of?

- a) $\mathcal{O}(N^3)$
- b) $\mathcal{O}(N \log N)$
- c) $\mathcal{O}(N^{\log_4 5})$
- d) $\mathcal{O}(N^{\log_2 5})$
- e) $\mathcal{O}(N^2)$

15. What is the edit distance between words 'ankara' and 'anarok'?

- a) 2
- b) 3
- c) 4
- d) 5
- e) 6

16. How many paths are there from cell (1, 1) to cell (5, 5) in a rectangular matrix, if the only possible moves are going right and going down?

- a) 15
- b) 20
- c) 35
- d) 45
- e) 70

17. What is the size of the longest increasing subsequence for 9, 21, 8, 32, 20, 49, 40, 59, 79?

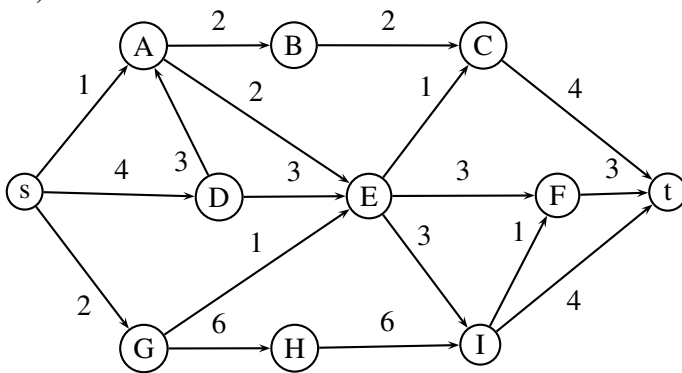
- a) 3
- b) 4
- c) 5
- d) 6
- e) 7

18. Given the matrices A, B, C, D with sizes 1×2 , 2×10 , 10×3 and 3×5 , which of the following order of multiplication requires minimum multiplications?

- a) $Ax((BxC)xD)$
- b) $(AxB)x(CxD)$
- c) $((AxB)xC)xD$
- d) $(Ax(BxC))xD$
- e) $Ax(Bx(CxD))$

19. Which of the following is an example P problem?

- a) 3SAT
- b) Traveling Salesman Problem
- c) Eulerian Path
- d) Integer Linear Programming
- e) Rudrata Path



20. In the graph above, what is the maximum flow from s to t ?

- a) 3
- b) 4
- c) 5
- d) 6
- e) 7

21. If we try to construct all possible subsets of an n element set via backtracking, how many candidates will be generated per construct_candidates method call?

- a) n
- b) 1
- c) 2^n
- d) 3
- e) 2

22. How can we convert multi-source / multi-sink problem to single-source / single-sink problem?

- a) Connect each source directly to each sink.
- b) Create a super source, connect super source to all sources, then connect each source directly to each sink.
- c) Create a super sink, connect each source to this super sink, then connect each sink to this super sink.
- d) Create a super source and super sink, connect super source to all sources, then connect all sinks to the super sink.
- e) Any one of the above

23. Let say we are constructing all numbers with 4 digits via backtracking, and let also say that we are at the candidate 3456, what will be the next candidate?

- a) 3446
- b) 2346
- c) 2556
- d) 3457
- e) 3453

24. Is $P \neq NP$?

- a) Yes
- b) No
- c) Unknown
- d) Can not be proven
- e) None of the above

25. What is the time complexity of the brute force search of the 3-SAT Problem with N variables?

- a) 3^N
- b) 2^N
- c) $N!$
- d) $\binom{N}{N/2}$
- e) N^2